Transparent Grant & Fund Tracking System

Built on Algorand Blockchain

Product Requirements Document • Hackathon 2024

1. Project Overview

A blockchain-powered platform that makes grant fund management completely transparent, automated, and tamper-proof. Built on Algorand, every rupee is tracked from sponsor to student — with zero human middlemen for fund release decisions.

"We didn't just track funds — we made honesty the only option."

Attribute	Details
Project Name	ChainGrant — Transparent Fund Tracker
Blockchain	Algorand (Testnet for Hackathon)
Difficulty	Intermediate
Estimated Build Time	18 hours (+ 6 hours polish & pitch)
Target Users	Students, Sponsors, College Admin, Public
Frontend	React
Backend	Node.js / Python
Smart Contracts	PyTeal / ARC4 on Algorand
Wallet	Pera Wallet

2. Problem Statement

Traditional grant management in educational institutions suffers from three critical failures:

- Lack of transparency — sponsors have no real-time visibility into how funds are spent
- No accountability — teams can go silent after receiving funds with no consequences
- Manual approvals — fund release depends on humans who can be slow, biased, or corrupt
- No verifiable proof — milestone completion is based on self-reporting with no immutable evidence
- Single-sponsor limitation — pooling funds from multiple sources is messy and untracked

ChainGrant eliminates all five problems by encoding the rules into smart contracts that execute automatically — no trust required.

3. Goals & Success Metrics

3.1 Primary Goals

- Enable milestone-based automated fund release via smart contracts
- Provide cryptographic proof of work and spending for every grant
- Allow multiple sponsors to pool funds with individual tracking
- Create a public, zero-login audit wall for complete transparency
- Build a reputation system via on-chain NFT badges

3.2 Success Metrics (Hackathon Demo)

Metric	Target
Live Algorand transactions demonstrated	5+
Features working end-to-end	All 6 unique features
Public transparency wall accessible	Yes — shareable URL
NFT minted successfully	Yes — on Algorand testnet
Dead Man's Switch simulation	Demonstrated in demo

4. Unique Features (The 6 Differentiators)

4.1 Multi-Sponsor Pooling

Multiple sponsors can contribute to a single grant like a crowdfunding campaign. Each contribution is tracked as a separate on-chain transaction, so every sponsor can independently verify their money's journey.

- Each sponsor gets their own wallet-linked contribution record
- Partial refunds possible if grant is cancelled
- Sponsors can set per-contribution conditions

4.2 On-Chain Proof of Work

Teams don't just say they completed a milestone — they submit a SHA-256 hash of their deliverable (code repo link, report PDF, prototype video URL) that is stored permanently on Algorand.

- Hash is computed client-side and anchored on-chain
- Provides a tamper-proof timestamp — stronger than any email or Google Doc
- Anyone can re-hash the original file and verify it matches

"Anyone, anywhere, forever can verify this work existed on this date."

4.3 Proof of Spend Receipt System

After funds are released, the student team uploads a spending receipt (photo or link). This receipt is hashed and stored on Algorand — nobody can alter or fake it retroactively.

- Receipt hash stored as Algorand Note field on the fund release transaction
- Linked directly to the milestone it corresponds to
- Creates a complete, unbreakable chain: Grant → Milestone → Spend

4.4 Dead Man's Switch for Stale Grants

If a team goes silent for 30 days after receiving funds with no milestone update, the smart contract automatically refunds the sponsor. No chasing, no emails, no human intervention required.

- Configurable inactivity window (default: 30 days)
- Auto-refund triggers on-chain with full audit trail
- Teams receive warning notifications at 20-day mark

"The blockchain holds teams accountable even when humans don't."

4.5 Public Transparency Wall

A no-login-required public page where anyone — parents, college admin, journalists, auditors — can see exactly where every rupee went, searchable by project name or sponsor.

- Single shareable URL per grant shows complete lifecycle
- Live data pulled directly from Algorand blockchain

- Searchable by project, sponsor, college, or date
- No account needed — open to the entire internet

4.6 Reputation NFT for Teams & Sponsors

When a grant is fully completed successfully, both the student team AND the sponsor receive a non-transferable NFT badge (Algorand Standard Asset) recording their track record on-chain.

- Non-transferable — reputation cannot be bought or sold
- Sponsors with more NFTs attract better student teams
- Teams with more NFTs get priority access to future grants
- Creates a self-reinforcing ecosystem of accountability

"Your integrity follows you on-chain. Good actors get rewarded. Bad actors can't hide."

5. User Roles & Journeys

Role	Key Actions	Blockchain Interaction
Sponsor	Create grant, pool funds, approve milestones	Wallet sign, fund lock, refund receive
Student Team	Submit milestones, upload proof, upload receipts	Hash submission, NFT receive
College Admin	View all grants, override disputes	Read-only on-chain verification
Public / Auditor	View transparency wall, verify hashes	No wallet needed — read-only

6. Technical Architecture

6.1 Tech Stack

Layer	Technology	Purpose
Frontend	React	UI, wallet connection, dashboard
Wallet	Pera Wallet SDK	User authentication & signing
Smart Contracts	PyTeal / ARC4	Milestone logic, fund release, Dead Man's Switch

Blockchain	Algorand Testnet	Immutable ledger, NFT minting
Hashing	SHA-256 (built-in JS/Python)	Proof of Work & Spend receipts
NFTs	Algorand Standard Assets (ASA)	Reputation badges
Backend (optional)	Node.js or Python	Off-chain metadata storage
Database (optional)	SQLite / Firebase	Non-critical metadata

6.2 Smart Contract Logic Flow

Step	Action	Who Triggers	On-Chain?
1	Grant created with milestone targets & amounts	Sponsor	Yes
2	Additional sponsors pool funds	Sponsor(s)	Yes
3	Funds locked in escrow smart contract	Automatic	Yes
4	Team submits milestone + SHA-256 hash	Student Team	Yes
5	Sponsor reviews and approves milestone	Sponsor	Yes
6	Funds auto-released to team wallet	Smart Contract	Yes
7	Team uploads spend receipt (hashed)	Student Team	Yes
8	30 days silence → auto-refund trigger	Smart Contract	Yes
9	Grant complete → NFT minted for both parties	Smart Contract	Yes

7. 24-Hour Build Plan

Hours	Task	Priority
0 – 3	Algorand basics: testnet setup, Pera Wallet, send first transaction	MUST
3 – 7	Smart contract: grant creation, milestone config, fund escrow	MUST
7 – 10	Smart contract: fund release + Dead Man's Switch logic	MUST

10 – 14	React frontend: grant creation form + milestone tracker	MUST
14 – 16	SHA-256 hashing for Proof of Work + Proof of Spend receipts	MUST
16 – 18	Multi-sponsor pooling UI + wallet integration	HIGH
18 – 20	Public Transparency Wall (read-only public page)	HIGH
20 – 22	Reputation NFT minting (Algorand ASA)	MEDIUM
22 – 24	Polish UI, test all flows, prepare demo script & pitch	MUST

8. Demo Script (Practice This)

This is the exact flow to demonstrate during judging. Each step should take under 60 seconds.

1. Two sponsors pool ₹10,000 into a grant — show both contributions tracked separately on-chain
2. Student team submits Milestone 1 + SHA-256 hash of their code repo — "This proof exists forever on Algorand"
3. Sponsor approves → funds auto-release to team wallet — show live Algorand transaction
4. Team uploads spend receipt → hashed and stored — "No one can alter this"
5. Open the Public Transparency Wall — paste the URL in judge chat, they verify it themselves
6. Simulate 30-day silence → show Dead Man's Switch auto-refund to sponsor
7. Grant complete → Reputation NFT minted for team and sponsor — show on Algorand explorer

9. Winning Pitch Lines

Use these during your presentation:

- **"Banks use vaults to protect money. We use code. Unlike a bank manager — our code cannot be bribed, pressured, or fired."**
- **"We didn't just add transparency. We made accountability automatic."**
- **"Every rupee has a story on our platform. And that story is written in permanent ink on the blockchain."**
- **"Sponsors don't need to trust student teams. They just need to trust math."**

10. Learning Resources (Zero to Algorand in 3 Hours)

Resource	What You'll Learn	Time
developer.algorand.org	Algorand basics, accounts, transactions	1 hr
github.com/algorand/sandbox	Local testnet setup	30 min
Pera Wallet docs	Wallet connection in React	30 min
PyTeal docs / ARC4 examples	Smart contract basics	1 hr
Algorand Testnet Dispenser	Get free ALGO for testing	5 min

Built with □ for the Algorand Hackathon

Making honesty the only option — one block at a time.